

Methods and Simulation Results for Bayesian Inference from Repeated Size Profiles

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1 Methods

1.1 Inferential targets

The simulation study distinguished four inferential targets:

1. individual-level survival, growth, and reproductive output as functions of individual size;
2. population-effective transition rates after averaging over unobserved attributes and demographic stochasticity;
3. functionals of the fitted projection kernel, including the dominant eigenvalue; and
4. prediction of future size profiles.

The individual-level vital rates were the primary targets for recovery experiments. This distinction was necessary because a model may accurately predict profile dynamics while estimating a population-effective transition operator that differs from the individual generating rates.

1.2 Individual-based population generator

At census time t , individual i had size x_{it} , age a_{it} , sex q_i , and optional persistent latent quality u_i . Survival was generated as an individual Bernoulli event,

$$S_{it} \sim \text{Bernoulli} \{s(x_{it}, a_{it}, q_i, u_i, E_t, N_t)\}, \quad (1)$$

where E_t denotes an environmental covariate and N_t denotes population size. Conditional on survival, next-census size was generated from a normal growth distribution,

$$x_{i,t+1} \mid S_{it} = 1 \sim \mathcal{N} \{m_g(x_{it}, a_{it}, q_i, u_i, E_t, N_t), \sigma_g^2(x_{it}, a_{it}, q_i, u_i, E_t, N_t)\}. \quad (2)$$

The number of realized recruits entering the next census was generated separately,

$$R_{it} \sim \text{Poisson} \{r(x_{it}, a_{it}, q_i, u_i, E_t, N_t)\}, \quad (3)$$

$$x_{\text{recruit},t+1} \sim \mathcal{N} \{m_b(x_{it}, E_t), \sigma_b^2\}. \quad (4)$$

Survival, growth, and reproduction were therefore separate stochastic individual events. The simulated population did not evolve by deterministic application of the recovery IPM.

For the initial broad experiments, the generating rates were

$$\text{logit}\{s(x)\} = -2.2 + 0.032x, \quad (5)$$

$$m_g(x) = \max(x, x + 18 - 0.12x), \quad \sigma_g = 4.5, \quad (6)$$

$$r_i(x) = I(q_i = \text{female}) \exp(-7 + 0.075x), \quad (7)$$

$$m_b(x) = 43 + 0.02x, \quad \sigma_b = 3.5. \quad (8)$$

The corresponding population-average recruitment rate incorporated the one-half expected female proportion. In later multi-truth experiments, the generator was parameterized directly from the recovery-model coefficients, with female reproductive output doubled so that its population average matched the declared recruitment curve.

1.3 Size-profile observation process

Individual identities were retained by the generator but discarded before profile-only analysis. At each census, individuals were independently sampled with profile detection probability $q_t(x)$, optionally with size measurement error. Observed sizes were grouped into equally spaced bins with centers x_j , width Δx , and bounds $(\ell_j, u_j]$. The baseline experiments used 5-mm bins.

The general observation intensity was

$$\lambda_t(x) = e_t q_t(x) \mu_t(x), \quad (9)$$

where $\mu_t(x)$ is latent population intensity and e_t is survey effort. Most experiments used known constant effort and size-independent detection. Conditional on the initial observed profile, subsequent bin counts were modeled using either

$$Y_{tj} \sim \text{Poisson}(M_{tj}) \quad (10)$$

or a Gamma–Poisson model,

$$\Lambda_{tj} \mid M_{tj}, \phi \sim \text{Gamma}(\phi, \phi/M_{tj}), \quad (11)$$

$$Y_{tj} \mid \Lambda_{tj} \sim \text{Poisson}(\Lambda_{tj}), \quad (12)$$

which yields a negative-binomial marginal distribution with mean M_{tj} and precision ϕ .

1.4 Positive inverse IPM

The recovery model propagated latent size intensity through a decomposed kernel,

$$\mu_{t+1}(x') = \int K(x', x) \mu_t(x) dx, \quad (13)$$

$$K(x', x) = P(x', x) + F(x', x), \quad (14)$$

$$P(x', x) = s(x)g(x' \mid x), \quad (15)$$

$$F(x', x) = r(x)b(x' \mid x). \quad (16)$$

The conditional growth and recruit-size densities were normalized:

$$\int g(x' \mid x) dx' = 1, \quad \int b(x' \mid x) dx' = 1. \quad (17)$$

Consequently, column integrals of P and F corresponded to survival and recruitment intensity. This normalization removed arbitrary scaling within each component, although it did not guarantee identification of the decomposition of K into P and F .

Vital-rate coefficients were centered at 80 mm, with $z(x) = (x - 80)/20$:

$$\text{logit}\{s(x)\} = \alpha_s + \beta_s z(x), \quad (18)$$

$$m_g(x) - x = \max\{0, \alpha_g + \beta_g z(x)\}, \quad (19)$$

$$\sigma_g = \exp(\eta_g), \quad (20)$$

$$\log r(x) = \alpha_r + \beta_r z(x), \quad (21)$$

$$m_b(x) = \alpha_b + \beta_b z(x), \quad (22)$$

$$\sigma_b = \exp(\eta_b). \quad (23)$$

The baseline model therefore estimated ten demographic coefficients.

1.5 Numerical integration

The continuous operator was discretized on a regular size mesh:

$$\boldsymbol{\mu}_{t+1} = \Delta x \mathbf{K} \boldsymbol{\mu}_t. \quad (24)$$

All kernel elements and projected intensities remained nonnegative. Conditional normal transition probabilities were integrated over target-size bins,

$$\Pr(\ell_j < X' \leq u_j \mid x) = \Phi\left\{\frac{u_j - m(x)}{\sigma}\right\} - \Phi\left\{\frac{\ell_j - m(x)}{\sigma}\right\}, \quad (25)$$

rather than evaluating normal densities only at bin centers. The resulting bin probabilities were normalized over the retained mesh. This correction was important when the 5-mm bins were wide relative to recruit-size variation.

1.6 Prior distributions

The broad Monte Carlo study used independent weakly informative normal priors on transformed coefficients. Selected induced 95% prior intervals were approximately:

Quantity	Prior median	Approximate 95% interval
Survival at 80 mm	0.60	0.07–0.97
Growth increment at 80 mm	8.0	0–19.8 mm
Growth SD	5.0	1.3–19.6 mm
Recruitment at 80 mm	0.20	0.01–3.78
Recruit mean size at 80 mm	45.0	25.5–64.6 mm
Recruit-size SD	4.0	1.0–15.8 mm

These priors were diffuse on biologically meaningful scales but retained structural information through the logistic and log links, parametric curve forms, normalized transition distributions, and nonnegative growth constraint. The Gamma–Poisson process precision received $\log \phi \sim \mathcal{N}\{\log(50), 2^2\}$.

1.7 External fecundity information

Auxiliary information was represented as an independent fecundity study. For an external study of n_e fish, standardized parent sizes z_i were balanced from -1.5 to 1.5 , and realized offspring counts were generated as

$$C_i \sim \text{Poisson} \left[\exp\{\alpha_r^{(e)} + \beta_r^{(e)} z_i\} \right]. \quad (26)$$

The external study was analyzed using a Poisson regression with the diffuse fecundity prior. A bivariate normal Laplace approximation to its posterior, including covariance between recruitment level and slope, became the external-study prior for the profile analysis.

Under full borrowing, this external posterior directly replaced the diffuse prior for (α_r, β_r) . Under robust borrowing, the prior was

$$\pi_{\text{robust}}(\alpha_r, \beta_r) = w \pi_e(\alpha_r, \beta_r) + (1 - w) \pi_0(\alpha_r, \beta_r), \quad w = 0.8, \quad (27)$$

where π_e denotes the external-study posterior approximation and π_0 denotes the diffuse fecundity prior.

Because the robust posterior could be genuinely multimodal, it was not approximated by a single Gaussian mode. Separate Laplace approximations were fitted under the external and diffuse components. Their posterior component weights were estimated from Laplace marginal-evidence approximations, and posterior draws were sampled from the resulting two-component mixture.

1.8 Posterior computation and approximation audit

Broad simulation grids used maximum a posteriori optimization and Gaussian Laplace posterior approximations. Non-positive-definite posterior Hessians were retained as approximation-failure records and excluded from scientific coverage summaries. Six representative datasets were also fitted using four adaptive-Metropolis chains. Across this audit, the median absolute difference between Laplace and MCMC posterior means was 0.106 MCMC posterior standard deviations, the median Laplace-to-MCMC posterior-SD ratio was 0.988, parameter coverage agreement was 100%, maximum split $\hat{R}$ was 1.063, and minimum approximate effective sample size was 66.

1.9 Simulation experiments

1.9.1 Broad design grid

The first broad grid varied allocation of a fixed 12-transition budget, initial population size, profile detection, initial size-profile shape, and recovery likelihood (Table 1).

Table 1: Factors in the broad Monte Carlo design.

Factor	Levels
Transition design	1×12 , 3×4 , 6×2
Initial population per trajectory	500, 2500
Fraction observed per annual profile	0.25, 1.00
Initial profile	baseline, small-fish cohort pulse
Independent replicates	10
Recovery likelihood	Poisson, Gamma-Poisson

This grid produced 240 independently simulated datasets and 480 Bayesian fits. The fixed transition budget separated the benefit of longer within-population follow-up from replication among independent population trajectories.

1.9.2 Auxiliary-fecundity experiment

The targeted auxiliary experiment used the 3×4 design, an initial population of 500 fish, and 25% profile detection. Thirty fresh profile datasets were independently generated. Each was analyzed using profiles alone; correctly transportable external fecundity studies of 50, 200, or 800 fish; an 800-fish biased external study with full borrowing; and an 800-fish biased external study with robust borrowing. Analyses were paired within profile dataset.

1.9.3 Transportability-boundary experiment

The transportability experiment considered three biological truths: baseline rates; survival at 80 mm reduced from 0.589 to 0.400; and recruitment slope per 20 mm reduced from 1.50 to 0.60. For each truth, ten fresh 3×4 profile datasets were generated. Each dataset was paired with an 800-fish external fecundity study.

Transport bias jointly shifted external-study recruitment level and slope:

$$\alpha_r^{(e)} = \alpha_r + 0.50d, \quad (28)$$

$$\beta_r^{(e)} = \beta_r - 0.30d, \quad (29)$$

where $d \in \{0, 0.25, 0.50, 0.75, 1.00, 1.50\}$. At $d = 0.25$, the external recruitment curve was approximately 27% higher at 50 mm, 13% higher at 80 mm, and 1% higher at 110 mm. Profiles-only, full-borrowing, and robust-borrowing analyses were compared on the same simulated datasets.

1.9.4 Paired profile and individual-data benchmark

The paired-data benchmark generated 200 independent datasets, each containing three populations with four annual transitions, 500 initial fish per population, 25% profile detection, and 35% known capture probability. Six analyses were fitted to each dataset: Poisson profiles only, Gamma–Poisson profiles only, capture–recapture transitions only, Poisson integrated profiles and capture–recapture, Gamma–Poisson integrated profiles and capture–recapture, and a complete-history oracle.

Conditional on capture of individual i in year t , consecutive recapture was modeled as

$$C_{i,t+1} \mid C_{it} = 1, x_{it} \sim \text{Bernoulli}\{s(x_{it})q\}, \quad (30)$$

where $q = 0.35$ was known. If recaptured, next-year size was modeled using the same conditional growth distribution as the inverse IPM. These data therefore informed survival and growth but supplied no reproductive or parent–offspring information. The oracle observed every individual survival outcome, growth transition, offspring count including zeros, parent–offspring link, and recruit size.

1.10 Performance measures

Performance measures included posterior bias, root mean squared error (RMSE), 95% interval width and coverage, and approximation diagnostics. Recruitment curve RMSE was evaluated at 50, 80,

and 110 mm. Auxiliary-study analyses were compared with profiles-only analyses using paired RMSE ratios. Bootstrap intervals resampled paired datasets. For the exploratory transportability boundary, borrowing was called safe only when the 95% bootstrap interval for the paired RMSE ratio was below one and coverage did not fall below profiles-only coverage.

2 Simulation Results

2.1 Numerical integration and initial identifiability

Evaluating conditional normal densities at 5-mm bin centers produced an artificial recruit-variance bias when the true recruit-size SD was 3.5 mm. Integrating normal probabilities over target bins removed this artifact. Numerical construction of the kernel was therefore consequential for inference rather than merely an implementation choice.

In the initial multi-year Bayesian experiment, repeated profiles accurately recovered survival probabilities and recruit-size means over representative sizes. With 12 transitions, estimated survival probabilities at 50, 80, and 110 mm were approximately 0.357, 0.589, and 0.787, compared with generating values 0.354, 0.589, and 0.789. In contrast, recruitment level, recruitment slope, some growth quantities, and the dominant eigenvalue were biased and overconfident. Recruitment level and slope had posterior correlations near -0.99 in short series. Additional years increased precision but could concentrate the posterior around a population-effective pseudo-truth that differed from individual generating rates.

2.2 Broad Monte Carlo recovery

The broad grid completed without fit failures; 98.96% of posterior Hessians were positive definite. Five non-positive-definite Poisson approximations were excluded from scientific coverage summaries.

Table 2: Mean 95% interval coverage by allocation of the 12-transition budget.

Design	Poisson parameters	Poisson derived	Gamma-Poisson parameters	Gamma-Poisson derived
1×12	0.734	0.701	0.764	0.718
3×4	0.795	0.775	0.818	0.818
6×2	0.795	0.813	0.816	0.836

Replicated trajectories outperformed one long trajectory on average (Table 2). However, very short trajectories retained recruitment-level/slope confounding. No design achieved nominal 95% coverage across all parameters and derived quantities.



Figure 1: Mean interval coverage across the broad simulation grid. Replicated trajectories generally improved recovery relative to one long trajectory, but coverage remained below nominal levels.

Increasing population size and profile completeness provided only modest improvement. Under Poisson recovery, mean parameter coverage increased from 0.755 for $N = 500$ with 25% detection to 0.795 for $N = 2500$ with complete profiles. The dominant limitation was decomposition of profile change into demographic mechanisms rather than uncertainty about annual profile shapes.

Table 3: Selected Gamma–Poisson coverage rates across the broad grid.

Quantity	Coverage
Recruit-size SD	0.958
Recruit mean at 110 mm	0.929
Recruit mean at 80 mm	0.908
Growth increment at 50 mm	0.892
Survival at 50 mm	0.879
Survival at 80 mm	0.829
Growth SD	0.754
Dominant eigenvalue	0.617
Recruitment at 80 mm	0.571

The size distribution of recruits was recoverable, whereas recruitment intensity was not (Table 3 and Figure 2). Independent Gamma–Poisson overdispersion modestly improved average coverage but did not resolve the primary structural confounding.

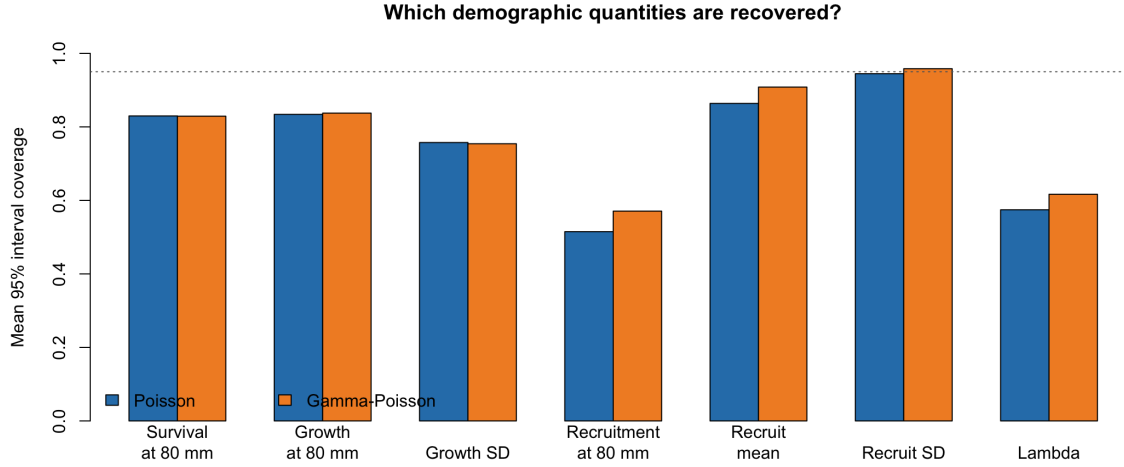


Figure 2: Coverage varied substantially among demographic quantities. Recruit-size distributions, survival, and mean growth were more reliably recovered than recruitment intensity and the dominant eigenvalue.

Applying the same small-fish cohort pulse to all populations did not uniformly improve recovery and reduced coverage for the 3×4 design. Useful perturbations must distinguish competing demographic mechanisms rather than simply add variation.

2.3 Moderate prior information

Moderately informative external priors, with uncertainty twice that of the initial oracle priors, changed most posterior summaries only modestly. Across the targeted prior-information experiment, mean coverage was 0.747 under diffuse priors, 0.769 with survival and growth information, 0.742 with fecundity information, and 0.747 with information on all vital rates. Mean relative absolute error decreased from 0.179 under diffuse priors to 0.161 with fecundity information and 0.155 with information on all rates.

The concentrated profile likelihood frequently overwhelmed moderate external information, including along directions where it targeted a population-effective pseudo-truth. Prior information was most useful when it directly constrained the confounded mechanism. Recruit-size information did not identify recruitment intensity.

2.4 Targeted auxiliary fecundity data

The confirmatory auxiliary-data experiment completed 180 analyses with no fit failures. Two non-positive-definite approximations from one difficult dataset were excluded from corresponding biased-prior summaries.

Table 4: Confirmatory auxiliary-fecundity results across 30 fresh profile datasets.

Analysis	Recruitment-curve RMSE	RMSE ratio	Coverage at 80 mm
Profiles only	0.280	1.000	0.533
Correct study, 50 fish	0.190	0.679	0.433
Correct study, 200 fish	0.123	0.440	0.600
Correct study, 800 fish	0.084	0.299	0.867
Biased study, 800 fish, full borrowing	0.102	0.366	0.000
Biased study, 800 fish, robust borrowing	0.238	0.852	0.276

Correctly transportable auxiliary fecundity data improved recruitment recovery as the external sample increased (Table 4). Two hundred external fish reduced mean recruitment-curve RMSE by more than half. Eight hundred external fish reduced RMSE by approximately 70% and increased recruitment-at-80 coverage from 53% to 87%.

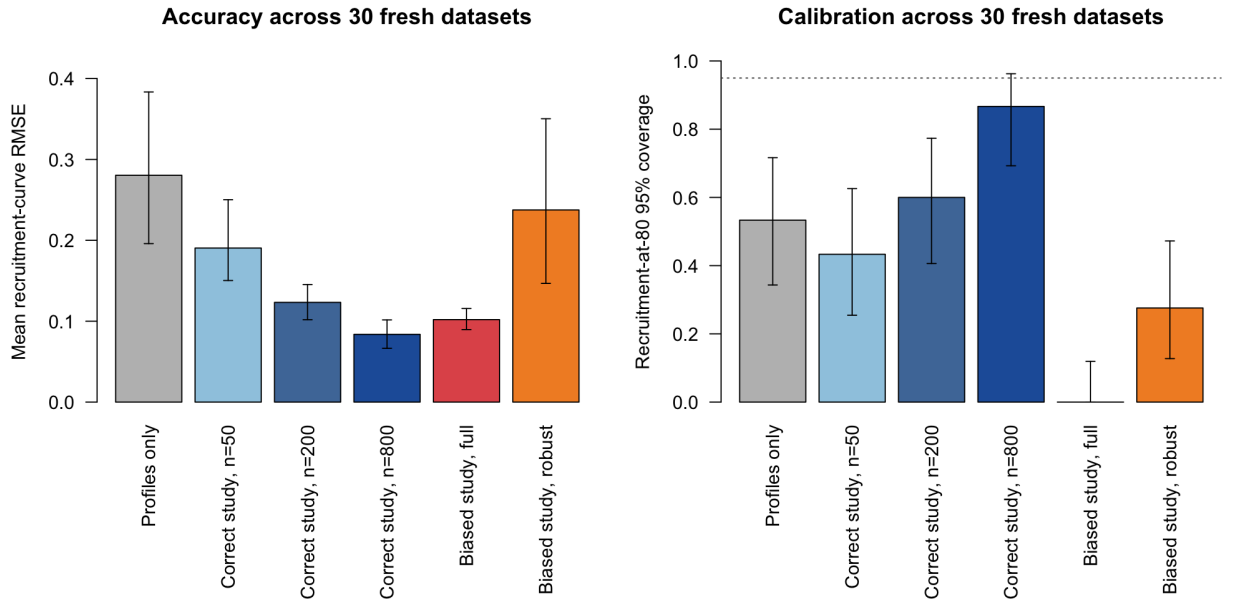


Figure 3: Confirmatory performance of targeted auxiliary fecundity data. Correctly transportable external data improved accuracy and eventually coverage. A precise biased study produced low RMSE but zero coverage; robust borrowing reduced but did not eliminate the damage. Error bars are bootstrap intervals for mean RMSE and exact binomial intervals for coverage.

The paired bootstrap 95% intervals for RMSE ratios were 0.502–0.921 for 50 external fish, 0.306–0.654 for 200 external fish, and 0.206–0.444 for 800 external fish. The precise biased study retained low point-estimate RMSE while producing zero coverage. Robust borrowing raised coverage to 0.276 but sacrificed most of the point-estimate gain. Its mean posterior external-study component probability remained approximately 0.67, indicating that profiles often could not reject the non-transportable study.

2.5 Transportability boundary

The multi-truth transportability experiment completed 390 fits with no failures and 98.2% positive-definite Hessians. One difficult baseline dataset and its unmatched component fits were excluded from paired boundary summaries.

Under full borrowing, the smallest tested mismatch reduced recruitment-at-80 coverage below profiles-only coverage for every biological truth. Pooled across truths, point-estimate RMSE remained substantially below profiles-only RMSE even after interval calibration had collapsed (Table 5).

Table 5: Full borrowing pooled across biological truths.

Bias multiplier	RMSE ratio versus profiles only	Coverage at 80 mm
0.00	0.356	0.862
0.25	0.396	0.483
0.50	0.390	0.103
0.75	0.442	0.000
1.00	0.532	0.000
1.50	0.760	0.000

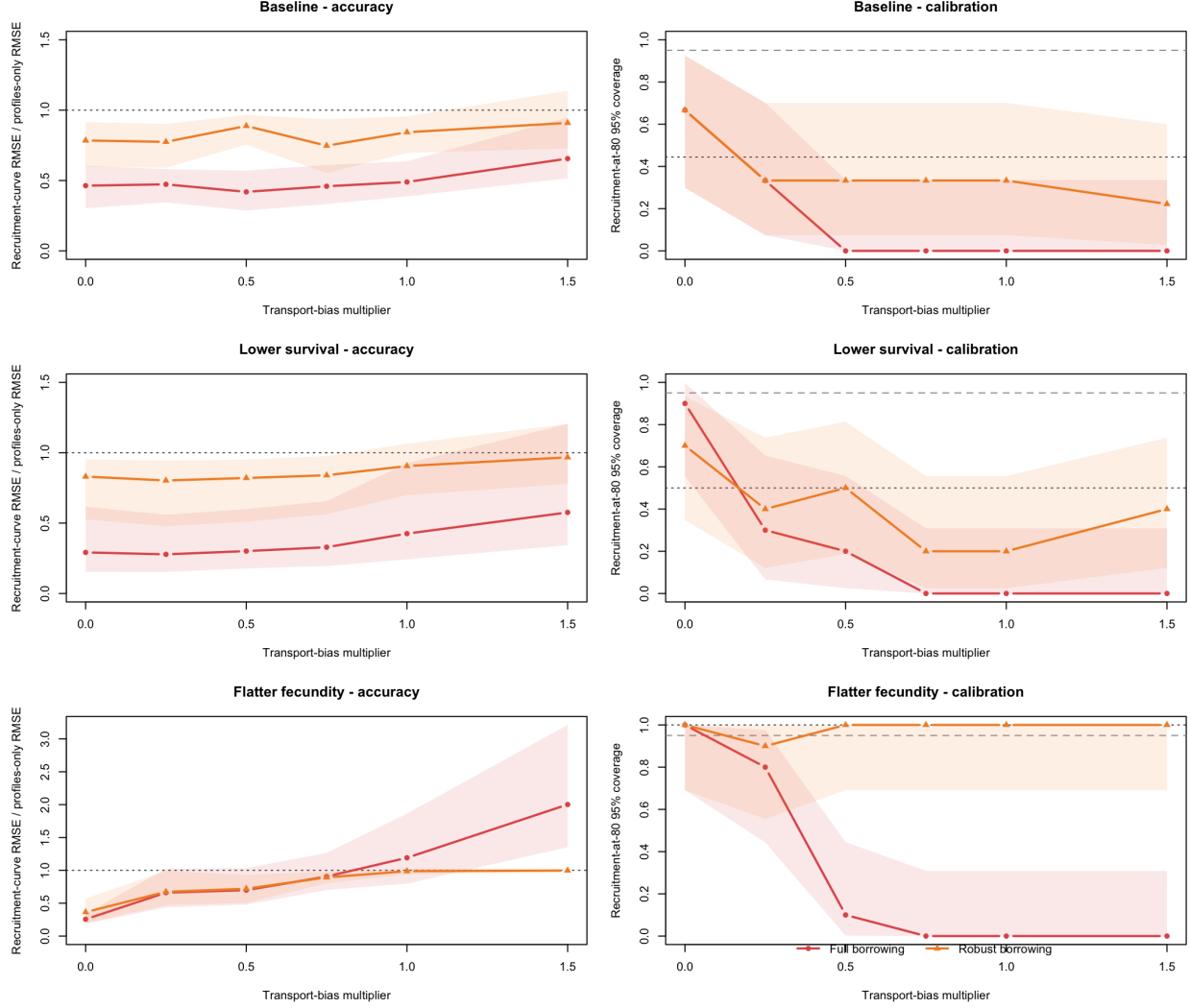


Figure 4: Accuracy and calibration as transport bias increased. Horizontal dotted lines show profiles-only performance; dashed lines show nominal 95% coverage. The safety of robust borrowing depended strongly on the generating biology.

Using the conservative definition that the paired RMSE-ratio interval must remain below one and coverage must not fall below profiles-only coverage, the largest tested safe full-borrowing bias was zero for all three truths. The largest tested safe robust-borrowing bias was zero under baseline biology, 0.50 under lower survival, and 0.75 under flatter fecundity. Because each boundary point used only nine or ten paired datasets, these boundaries should be treated as exploratory.

Robust borrowing rejected conflict only when profile dynamics were informative about that conflict. Under flatter fecundity, posterior external-study weight fell from approximately 0.91 with no bias to effectively zero by bias multiplier 1.5. Under baseline and lower-survival biology, external-study weight remained approximately 0.55–0.70 even under the largest tested bias (Figure 5).

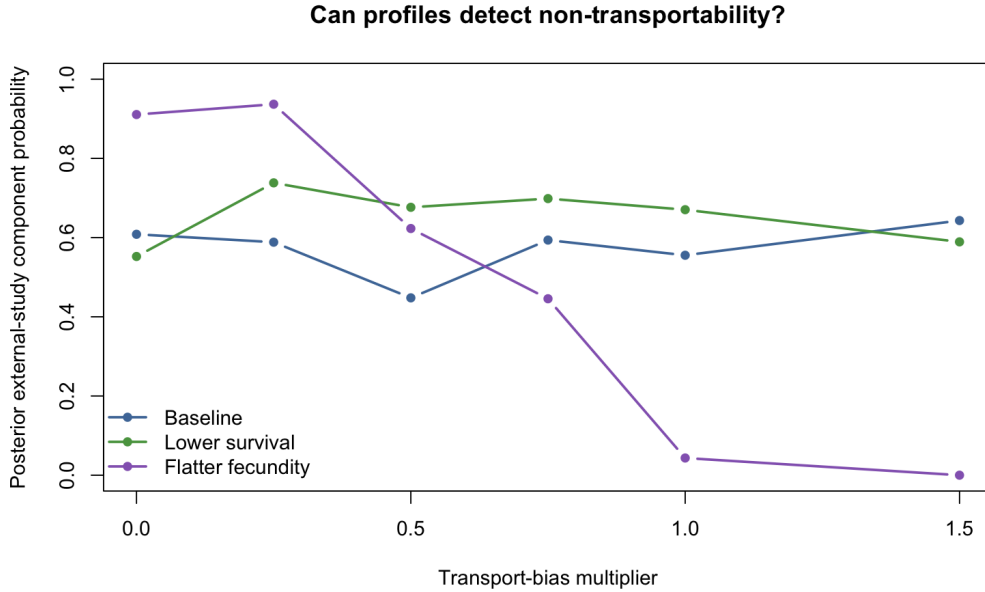


Figure 5: Mean posterior probability assigned to the external-study component under robust borrowing. Profiles detected non-transportability under flatter fecundity but not reliably under baseline or lower-survival biology.

2.6 Paired profile and individual-data benchmark

All 1,200 paired-data fits completed, and 1,191 (99.25%) had positive-definite posterior Hessians. Approximation failures occurred only under profile-only recovery: two Poisson and seven Gamma-Poisson fits. Across datasets, the mean numbers observed were 1,499 fish in size profiles, 2,100 capture records, and 272 consecutive recaptures.

Capture-recapture directly and efficiently identified survival and growth (Table 6). Its apparently perfect recruitment coverage did not indicate recovery: recruitment and recruit-size parameters contracted by less than 1% from their weak priors, and the resulting recruitment and population-growth intervals were extremely wide.

Table 6: Paired-data benchmark across 200 simulated datasets. Entries are empirical 95% interval coverage / RMSE of posterior means.

Analysis	Survival at 80 mm	Growth increment at 80 mm	Recruitment at 80 mm	Population growth
Poisson profiles	0.914 / 0.0267	0.889 / 0.904	0.495 / 0.143	0.737 / 0.0284
Gamma-Poisson profiles	0.948 / 0.0257	0.876 / 0.905	0.570 / 0.148	0.772 / 0.0299
Capture-recapture	0.970 / 0.0394	0.940 / 0.258	1.000 / 0.440	1.000 / 6.119
Poisson integrated	0.795 / 0.0232	0.935 / 0.285	0.485 / 0.148	0.715 / 0.0258
Gamma-Poisson integrated	0.850 / 0.0216	0.930 / 0.274	0.585 / 0.150	0.760 / 0.0264
Complete-history oracle	0.960 / 0.0098	0.950 / 0.102	0.930 / 0.009	0.905 / 0.0140

Integrated analyses improved survival point accuracy, but their intervals became too narrow. Survival-at-80 coverage fell from 0.97 under capture-recapture alone to 0.80 under Poisson integration and 0.85 under Gamma-Poisson integration, despite lower RMSE. Recruitment coverage remained poor. This pattern indicates that accurate individual transition information anchored survival and

growth while remaining discrepancies between observed and deterministically projected profiles were absorbed by other demographic parameters.

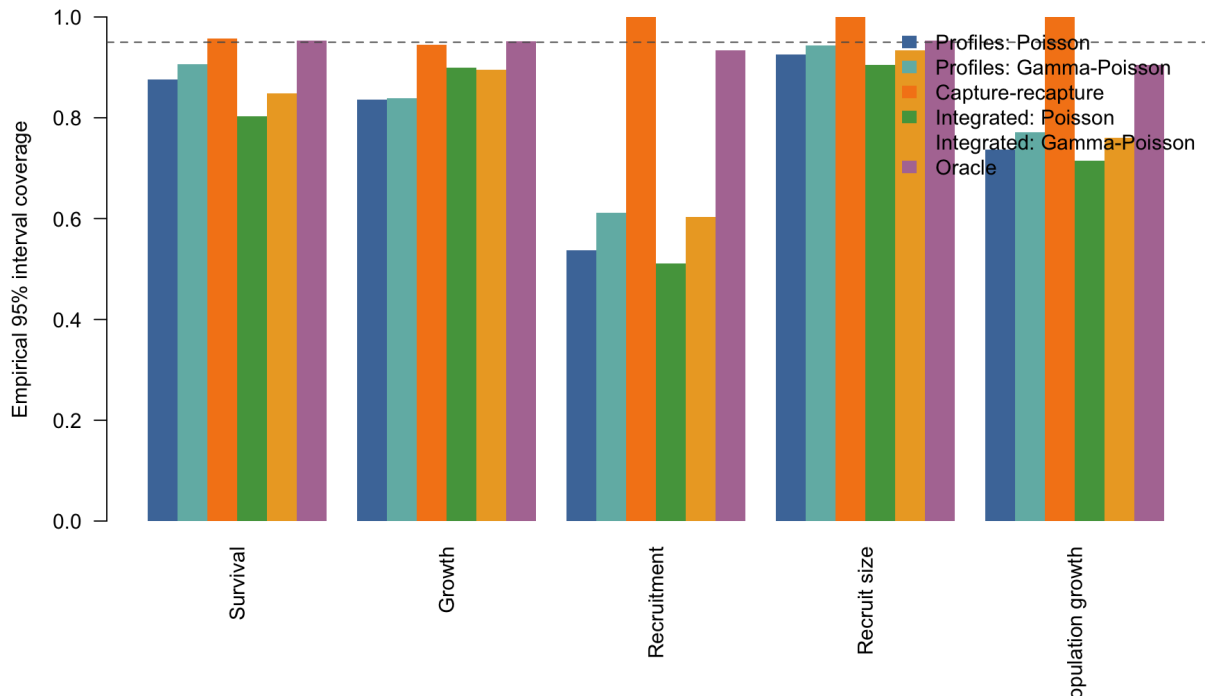


Figure 6: Empirical interval coverage in the paired-data benchmark. Perfect capture–recapture coverage for recruitment, recruit size, and population growth reflects prior-dominated wide intervals rather than data-based identification.

Gamma–Poisson overdispersion improved calibration but did not restore nominal coverage. For example, it increased integrated recruitment coverage from 0.51 to 0.60 and integrated survival coverage from 0.80 to 0.85. Independent bin-level overdispersion therefore absorbed some, but not all, of the structured discrepancy generated by finite individual populations.